

Formulae for D^0 Photoproduction Cross Sections diffractive-D0-UPC

This note collects the formulae actually implemented in the `diffractive-D0-UPC` code, for open-charm (D^0) photoproduction in ultra-peripheral Pb+Pb collisions in the color-dipole picture. Every equation is annotated with the source file and line where it is coded, so it can be checked directly against the implementation. Two processes are computed:

- **Exclusive:** $\gamma A \rightarrow c\bar{c} A$, coherent on the nucleus (no extra radiation).
- **Diffractive:** $\gamma A \rightarrow c\bar{c} g A$, with an extra radiated gluon and a rapidity gap ($x_{IP} < 0.1$).

Both share the same photon flux and kinematics, and both are followed by $c \rightarrow D^0$ fragmentation.

1 Kinematics

The observed D^0 transverse momentum p_{D^0} relates to the charm quark's transverse momentum p_c through the fragmentation variable $z_h \in [z_{\min}, z_{\max}] = [0.05, 1]$:

$$p_c = \frac{p_{D^0}}{z_h}, \quad m_T = \sqrt{p_c^2 + m_c^2} \quad (1)$$

`integrand.cpp:96,100`

The charm quark's light-cone plus-momentum is fixed by rapidity y , and the photon's plus-momentum q^+ is a Monte Carlo integration variable in $[p^+, q_{\max}^+]$:

$$p^+ = \frac{m_T}{\sqrt{2}} e^y, \quad z_1 = \frac{p^+}{q^+}, \quad z_2 = 1 - z_1 \quad (2)$$

`integrand.cpp:101,106,111--112`

The photon-nucleon invariant mass squared W^2 , the $c\bar{c}$ invariant mass squared $M_{q\bar{q}}^2$, and the pomeron momentum fraction x_{IP} (exclusive) are

$$W^2 = \sqrt{2} q^+ \sqrt{s}, \quad M_{q\bar{q}}^2 = \frac{p_c^2 + m_c^2}{z_1 z_2}, \quad x_{IP} = \frac{M_{q\bar{q}}^2}{W^2} \quad (3)$$

`integrand.cpp:116--118`, $\sqrt{s} = 5360$ GeV (Pb-Pb)

with the physical region restricted to $0 < x_{IP} \leq 0.1$ (diffractive requirement), and the dipole amplitude frozen for $x_{IP} < 0.01$: $x_{\text{dip}} = \min(x_{IP}, 0.01)$.

For the diffractive channel, an extra gluon of transverse momentum $k \leq p_c$ is radiated, carrying light-cone fraction x of the pomeron:

$$x = \frac{M_{q\bar{q}}^2}{x_{IP} W^2}, \quad \omega^2 = \frac{x}{1-x} k^2 \quad (4)$$

`integrand.cpp:233,237`

2 Photon flux

The equivalent-photon (Weizsäcker–Williams) flux density at impact parameter b is

$$n(q^+, b) = \frac{\alpha_{\text{em}} Z^2}{\pi^2} \frac{1}{\omega} \frac{\eta^2}{b^2} K_1(\eta)^2, \quad \omega = \frac{q^+}{\sqrt{2}}, \quad \eta = \frac{2\omega}{\sqrt{s}} m_N b \quad (5)$$

gamma_aa.cpp:20--30, $Z = 82$, $m_N = 0.938$ GeV

This is multiplied by two survival probabilities and integrated over impact parameter:

$$\Phi(b, q^+) = 2\pi b n(q^+, b) \Gamma_{AA}(b) P_{\text{no emd}}(b) \quad (6)$$

gamma_aa.cpp:32--45

- $\Gamma_{AA}(b)$: probability of no hadronic (strong) interaction between the two nuclei, a cubic-spline interpolation of a tabulated nuclear-overlap Glauber calculation (`data/Gamma_AA.dat`), $\Gamma_{AA} \rightarrow 1$ for $b \gtrsim 150$ fm.
- $P_{\text{no emd}}(b) = e^{-S/b^2}$, with $S = (17.4 \text{ fm})^2$: probability of no mutual electromagnetic dissociation (no extra neutron emission). The *channel* selects the neutron-tag class,

$$\text{channel} = \begin{cases} P_{\text{no emd}}(b) & \text{An0n (no extra neutrons on side A)} \\ P_{\text{no emd}}(b) [1 - P_{\text{no emd}}(b)] & \text{Xn0n (neutrons on one side only)} \end{cases} \quad (7)$$

gamma_aa.cpp:39--43, S : main.cpp:86

$G=1$ variant: setting $\Gamma_{AA}(b) \equiv 1$ removes the hadronic-survival suppression (used for the `_G1` comparison files).

3 Dipole amplitude

$N(r, x)$ is the imaginary part of the dipole scattering amplitude for a $q\bar{q}$ pair of transverse size r at Bjorken- x , obtained externally from Balitsky–Kovchegov (BK) evolution and read from a data grid (`AmplitudeLib::N`, `amplitudelib.hpp:71`) – either a fixed proton solution (`data/proton/mve.dat`) or a Glauber-averaged nuclear one at fixed b (`data/Pb/mve/glauber_mve_{\langle b \rangle}`).

4 Exclusive cross section

Define, for the two Bessel orders $\nu = 0, 1$,

$$I_\nu(p_c, m_c, x_{IP}) = \int_0^\infty dr r J_\nu(p_c r) K_\nu(m_c r) N(r, x_{IP}) \quad (8)$$

integrand.cpp:53--57

the unfragmented ($c\bar{c}$ -level) exclusive kernel is

$$g(p_c) = (z_1^2 + z_2^2) I_1^2 + I_0^2, \quad \frac{d\sigma_{c\bar{c}}}{d^2p_c dy} \propto z_1 m_c^2 g(p_c) \quad (9)$$

integrand.cpp:125--128

Folding in the photon flux (Eq. 6) and the $c \rightarrow D^0$ fragmentation (Sec. 6), the full Monte Carlo integral computed by `exclusiveCrossSection` is

$$\sigma_{\text{excl}}^{\text{MC}} = \int_{b_{\text{min}}}^{b_{\text{max}}} db \int_{p^+}^{q_{\text{max}}^+} dq^+ \int_{z_{\text{min}}}^{z_{\text{max}}} dz_h z_1 m_c^2 g(p_c) \Phi(b, q^+) \frac{D(z_h)}{z_h^2} \quad (10)$$

int_exclusive.cpp:20--74, integrand.cpp:87--131

and the physical differential cross section is

$$\boxed{\frac{d\sigma_{\text{excl}}}{dy d^2p_{D^0}} = \frac{\alpha_{\text{em}} N_c e_c^2 \sigma_0}{2\pi^2 m_c^2} \sigma_{\text{excl}}^{\text{MC}}} \quad (11)$$

python/combine_nucleus.py:17,85, $e_c = \frac{2}{3}$, $N_c = 3$, $\sigma_0 = 16.36$ mb

Nucleus target only: when the dipole amplitude already comes from a Glauber average at fixed b (as for Pb) and $\sigma_{\text{excl}}^{\text{MC}}$ already includes the b -integral of Eq. 10, the proton-dipole normalization σ_0 is *dropped* from Eq. 11 (it would double-count the target area, already supplied by the Glauber b -integral): $\sigma_0 \rightarrow 1$.
python/cross_section.py:120--131

5 Diffractive cross section

In addition to $\{r_1, r_2, q^+, b, z_h\}$, the diffractive channel integrates over the radiated gluon's momentum $k \in [0.01, p_c]$ GeV and the pomeron momentum fraction $x_{IP} \in [10^{-6}, 0.1]$ (both sampled log-uniformly). The “hard factor” and the gluon transverse-momentum-dependent piece are

$$H^2 = \frac{z_1}{(p_c^2 + m_c^2)^4} \left[(z_1^2 + z_2^2)(p_c^4 + m_c^4) + 2m_c^2 p_c^2 \right] \quad (12)$$

$$g_{\text{DTMD}}^2 = \omega^4 r_1 r_2 J_2(kr_1) K_2(\sqrt{x/(1-x)} k r_1) J_2(kr_2) K_2(\sqrt{x/(1-x)} k r_2) \quad (13)$$

integrand.cpp:241--249

so that the diffractive Monte Carlo integral is

$$\sigma_{\text{diff}}^{\text{MC}} = \int db dq^+ dz_h dx_{IP} dk dr_1 dr_2 2\pi k \Phi(b, q^+) N(r_1, x_{IP}) N(r_2, x_{IP}) \frac{H^2 g_{\text{DTMD}}^2}{1-x} \frac{D(z_h)}{z_h^2} \quad (14)$$

integrand.cpp:194--262, int_diffractive.cpp:14--55

and the physical cross section, now including the (running) strong coupling evaluated at the charm transverse mass, is

$$\boxed{\frac{d\sigma_{\text{diff}}}{dy d^2p_{D^0}} = \frac{\alpha_s(m_T) \alpha_{\text{em}} e_c^2 (N_c^2 - 1) \sigma_0}{8\pi^4} \sigma_{\text{diff}}^{\text{MC}}} \quad (15)$$

int_diffractive.cpp:145--151, python/combine_nucleus.py:18,88

with the same nucleus-target convention as above ($\sigma_0 \rightarrow 1$ when the b -Glauber integral is already included).

6 $c \rightarrow D^0$ fragmentation

The weight $D(z_h)/z_h^2$ in Eqs. 10,14 uses one of three fragmentation models (integrand.cpp:31--43):

BCFY (default). Braaten–Cheung–Fleming–Yuan, combining a pseudoscalar (D_P) and a vector (D_V) piece with the non-perturbative parameter $r = 0.1$:

$$D_c^{D^0}(z) = 0.168 D_P(z, r) + 0.39 \theta\left(\frac{m_{D^0}}{m_{D^{*0}}} - z\right) \frac{m_{D^{*0}}}{m_{D^0}} D_V\left(\frac{m_{D^{*0}}}{m_{D^0}} z, r\right) \quad (16)$$

fragmentation.cpp:9--63, $m_{D^0} = 1.8648$, $m_{D^{*0}} = 2.0067$ GeV with D_P, D_V the standard BCFY functional forms (degree-4 polynomials in z over $[1 - (1 - r)z]^{-6}$).

Kniesl–Kramer.

$$D(z) = N \frac{z(1-z)^2}{[(1-z)^2 + \epsilon z]^2}, \quad N = 0.694, \epsilon = 0.101 \quad (17)$$

`fragmentation.cpp:66--72`

LHAPDF. An external fragmentation-function grid, interpolated at scale $Q = (\text{scale factor}) \times m_{T,D^0}$ with $m_{T,D^0} = \sqrt{p_{D^0}^2 + m_c^2}$ (scale factor = 1 by default; $0.5 \times / 2 \times$ for the scale-uncertainty envelope), floored at $Q = m_c$. `main.cpp:111--128`

7 Running coupling α_s

One-loop running, used only in the diffractive prefactor (Eq. 15), evaluated at $\mu = m_T = \sqrt{p_c^2 + m_c^2}$ (charm-quark transverse mass):

$$\alpha_s(\mu) = \frac{\alpha_s(m_Z)}{1 + 2b_0 \alpha_s(m_Z) \ln(\mu/m_Z)}, \quad b_0 = \frac{33 - 2N_f}{12\pi} \quad (18)$$

`alphas_running.cpp`, $\alpha_s(m_Z) = 0.118$, $m_Z = 91.2 \text{ GeV}$, $N_f = 4$

8 Units

Raw results are in GeV^{-2} (natural units); converted to physical cross sections via $1 \text{ GeV}^{-2} = 0.3894 \text{ mb}$, and $d^2 p_{D^0} \rightarrow 2\pi p_{D^0} dp_{D^0}$ when plotted differential in p_{D^0} (not $p_{D^0}^2$). `python/cross_section.py:48--51,167--`

Note: the values in `files/inclusive/` (used for the $\Gamma=1$ vs. nominal-flux comparison) come from a separate, sibling project (`inclusive-D0-UPC`) with its own normalization convention and are not derived from the formulae above; ask if you'd like those documented as well.